

Subliminal Persona Transfer in Language Models

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next-level-personas

Abstract

Recent work has shown that language models transmit behavioral traits to student models through the statistical structure of generated data, independent of its semantic content (Cloud et al., 2025). Separately, persona vectors encode rich personality traits as linear directions in a model’s activation space (Chen et al., 2025). We ask whether these phenomena interact: can a multi-dimensional persona induced in a teacher’s activation space be transferred subliminally to a student fine-tuned only on the teacher’s number outputs? We construct teachers with personas (e.g., “evil”) induced via persona vectors, generate semantically unrelated number datasets, and fine-tune students on them. We evaluate transfer using behavioral probes and by testing whether the student’s residual stream shifts toward the original persona vector. Our work extends subliminal learning from scalar preferences to structured personas.

1 Introduction

Language models are increasingly trained on data generated by other language models, under the assumption that semantic filtering ensures safe student behavior. Cloud et al. (2025) challenge this, showing that behavioral traits leak through the statistical structure of generated data regardless of its content. No prior work has tested whether a full *persona* vector can be transferred subliminally. Persona vectors encode richer, more structured dispositions than a preference for a single word, and it is unclear whether the subliminal channel has sufficient bandwidth to carry them. We ask:

1. Can a misaligned persona encoded in a teacher’s activation space be transmitted to a student through semantically unrelated generated data?
2. Does the student’s residual stream shift in the direction of the original persona vector?

Answering these questions has direct safety implications: a compromised or steered model in a data pipeline could silently propagate its personality to downstream models, even through data that passes semantic filtering.

2 Approach

We follow the subliminal learning setup of Cloud et al. (2025), using Llama 3 as our base model family. Persona vectors are extracted following Chen et al. (2025): given contrastive prompt pairs eliciting opposite ends of a trait (e.g., evil vs. neutral), a steering vector is computed as the mean difference in residual stream activations. A teacher is steered by adding this vector to its residual stream and generates number sequences containing no explicit reference to the induced persona. A student of the same family is then fine-tuned on this data. We study two personas, *evil* and *sycophantic*. As baselines, we compare against students trained on numbers from an unsteered teacher (lower bound) and the steered teacher itself (upper bound). We evaluate transfer behaviorally using the free-form prompts of Betley et al. (2025) and quantitatively via cosine similarity between the student’s residual stream and the original steering vector.

3 Project Timeline

- **By May 24:** Set up pipeline. Persona vector extraction and teacher steering (Wiktorja); number dataset generation and filtering (Dominic); student fine-tuning infrastructure (Yasmine); evaluation harness and behavioral probes (Victor).
- **By June 7:** Activation-space analysis and cosine similarity measurements (Wiktorja, Yasmine); ablations across personas and model scales (Dominic, Victor); final report writing (all members).

References

- Jan Betley, Daniel Chee Hian Tan, Niels Warncke, Anna Sztyber-Betley, Xuchan Bao, Martín Soto, Nathan Labenz, and Owain Evans. 2025. [Emergent misalignment: Narrow finetuning can produce broadly misaligned LLMs](#). In *Proceedings of the 42nd International Conference on Machine Learning*, volume 267 of *PMLR*, pages 4043–4068.
- Runjin Chen, Andy Ardit, Henry Sleight, Owain Evans, and Jack Lindsey. 2025. [Persona vectors: Monitoring and controlling character traits in language models](#).
- Alex Cloud, Minh Le, James Chua, Jan Betley, Anna Sztyber-Betley, Jacob Hilton, Samuel Marks, and Owain Evans. 2025. [Subliminal learning: Language models transmit behavioral traits via hidden signals in data](#).